

SYNOPSIS
ON
Legal Assistant chatbot using Machine Learning
(Legal Compass)

Submitted in
Partial Fulfillment of requirements for the Award of Degree
of
Bachelor of Technology
In
Computer Science and Engineering
By

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Kunal (2301641640032)
Alekh Sachan (2301641640004)
Roshni Samanta (2301641640044)

Under the supervision of
Mr. Rajendra Kumar
(Associate Professor)



Pranveer Singh Institute of Technology.
Kanpur - Agra - Delhi National Highway - 19
Bhauti - Kanpur - 209305.
(Affiliated to Dr. A.P.J. Abdul Kalam Technical University)

1. Introduction

LegalCompass is an AI-powered platform designed to provide users with basic legal information. The project aims to bridge the gap between complex legal jargon and the general public's understanding by simplifying legal knowledge in an easily accessible format. The field of this project falls under LegalTech, which leverages technology to enhance the delivery of legal services and make legal information more transparent and widely available. The platform will feature automated AI responses to legal queries, making it a cost-effective and user-friendly tool for legal awareness.

The key technical terms associated with this project include AI-based knowledge processing, natural language processing (NLP), and legal information systems. These technologies help *LegalCompass* deliver accurate and contextually relevant legal information to users. By utilizing a futuristic and advanced website design, the project aims to create an interactive and intuitive user interface that enhances the user experience.

2. Project Objective

The primary objective of *LegalCompass* is to democratize access to basic legal knowledge by providing a user-friendly, AI-powered platform that allows users to get answers to common legal questions. By the end of the project, the platform should be able to:

- Provide users with accurate legal information on topics such as contracts, property, business, and criminal laws.
- Use natural language processing (NLP) to understand user queries and deliver appropriate responses.
- Offer a futuristic website design that enhances usability and interaction, ensuring accessibility on various devices.
- Help individuals, especially those who cannot afford legal services, to better understand their rights and responsibilities under the law.

The platform will empower end users by reducing their dependence on expensive legal consultations for basic queries. It will also improve legal awareness, making legal concepts more transparent and accessible to the general public.

3. Feasibility Study

The feasibility study assesses the practicality of implementing the *LegalCompass* platform, focusing on key areas:

- **Technical Feasibility:** The project uses proven technologies like natural language processing (NLP) and AI. The required infrastructure, such as cloud computing, is available and cost-effective.
- **Operational Feasibility:** With automated AI-driven responses, the platform will operate efficiently with minimal human intervention.
- **Economic Feasibility:** The project's economic viability will be ensured through:
 - **AdSense Revenue:** By integrating Google AdSense, the platform can generate revenue through ads.
 - **Minimal User Charges:** Users will pay small fees for premium features, such as advanced legal resources or personalized consultations.
 - **Premium Services:** Advanced legal consultations will be offered as paid services, creating an additional income stream.

These revenue streams will cover the platform's costs and ensure sustainability.

- **Schedule Feasibility:** The project is scheduled to run from 09-Sep-2024 to 30-Nov-2024, with well-defined phases for development and testing.
- **Legal Feasibility:** The platform will comply with privacy regulations like GDPR to protect user data.

Gantt Chart:

PROCESS	SEPTEMBER				OCTOBER				NOVEMBER			
	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4
Project Planning												
Requirement Analysis												
System & AI design												
AI model & Web Development												
Testing & Debugging												
Final Review & Deployment												

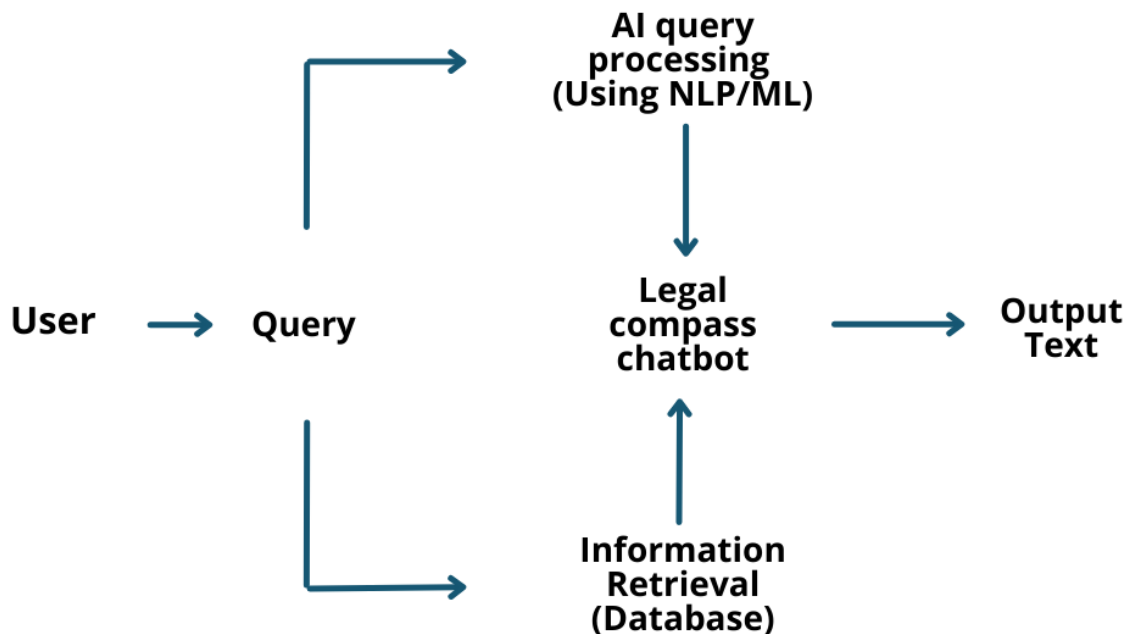
4. Methodology/Planning of Work

The development of LegalCompass will follow these key steps:

1. Requirement Analysis:
- Identify user needs and define system requirements.
2. System Design:
- Create architecture and Data Flow Diagrams (DFD) to illustrate information flow.

- Design the AI model and user interface.
- 3. AI & NLP Model Development:**
 - Build and train the AI model for legal query processing.
- 4. Website Development:**
 - Develop a responsive website with integrated AI functionality and AdSense.
- 5. Testing & Debugging:**
 - Test for functionality, accuracy, and fix bugs.
- 6. Deployment & Feedback:**
 - Launch the platform and gather user feedback for improvements.

Data Flow Diagram:



5. Tools/Technology Used

5.1 Minimum Hardware Requirements

- **CPU:** Intel i5 or equivalent (Quad-Core, 2.5 GHz or higher)
- **RAM:** 8 GB minimum (16 GB recommended for smoother performance)
- **GPU:** Integrated GPU or NVIDIA GTX 1050 (for AI model training)
- **HDD:** 256 GB SSD (512 GB recommended for faster access and storage)
- **Others:** Stable internet connection, basic input/output peripherals (keyboard, mouse)

5.2 Minimum Software Requirements

- **Operating System:** Windows 10 / Linux (Ubuntu 20.04) / macOS
- **Development Tools:**
 - **Python 3.9** (for AI & NLP model development)
 - **Flask 2.0** (for web framework and backend development)
 - **TensorFlow 2.0** or **PyTorch** (for AI and machine learning models)
 - **MySQL** or **SQLite** (for database management)
 - **HTML5/CSS3/JavaScript** (for front-end website development)
 - **Google AdSense API** (for ad integration)
 - **Git** (for version control)

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